



Spendify

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Executive Summary

Spendify is an AI-powered expense analysis and classification system that simplifies receipt-based financial tracking. The system enables users to upload receipts via Discord, processes them through Google Vertex AI's Document OCR, classifies line items using a structured multi-agent ADK pipeline, and stores the results in Firebase. A web dashboard (under development) provides visual summaries and spend forecasting using a regression model.

Originally initiated with *LangChain/Ollama* for classification, Spendify transitioned to a robust, explainable multi-agent pipeline using **Google Cloud's Agent Development Kit (ADK)**. The backend, built in Flask, integrates seamlessly with the Discord bot and dashboard. Firebase Firestore stores both raw and classified data. Key challenges included LLM inconsistencies, GCP ADK deployment, and cloud authentication. The final product successfully demonstrates an end-to-end automated receipt processing pipeline with future expansion plans for dashboard enhancements and AI chat integration.

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Introduction

Problem Statement

Manual receipt tracking is tedious, error-prone, and lacks intelligence. Users need a seamless way to digitize and analyze spending habits using AI.

Objective

Develop a receipt-to-dashboard pipeline that supports:

- Discord-based receipt upload
- OCR data extraction via Document AI from GCP Vertex AI Platform

- LLM/agent-based classification (GCP ADK)
- Spend prediction (regression model)
- Web dashboard for analytics

Milestone Alignment

- M1: Defined system scope, bot integration, API setup, and OCR planning.
- M2: Implemented LLM classification, OCR via Vertex AI, structured Firebase storage.
- M3: Replaced LLM with ADK agent pipeline; built validated end-to-end flow.
- M4: Demonstrated deployed infrastructure, API endpoints, and partial dashboard with testing validation.

System Architecture & Technical Design

Core Components

- Discord Bot: Receives image uploads.
- Flask API: Processes upload, sends images to OCR, returns predictions.
- Vertex AI (OCR): Extracts structured text.
- ADK Pipeline: Multi-agent system for item classification with validation.
- Firebase: Stores raw, structured, and summarized receipt data.

Technologies Used

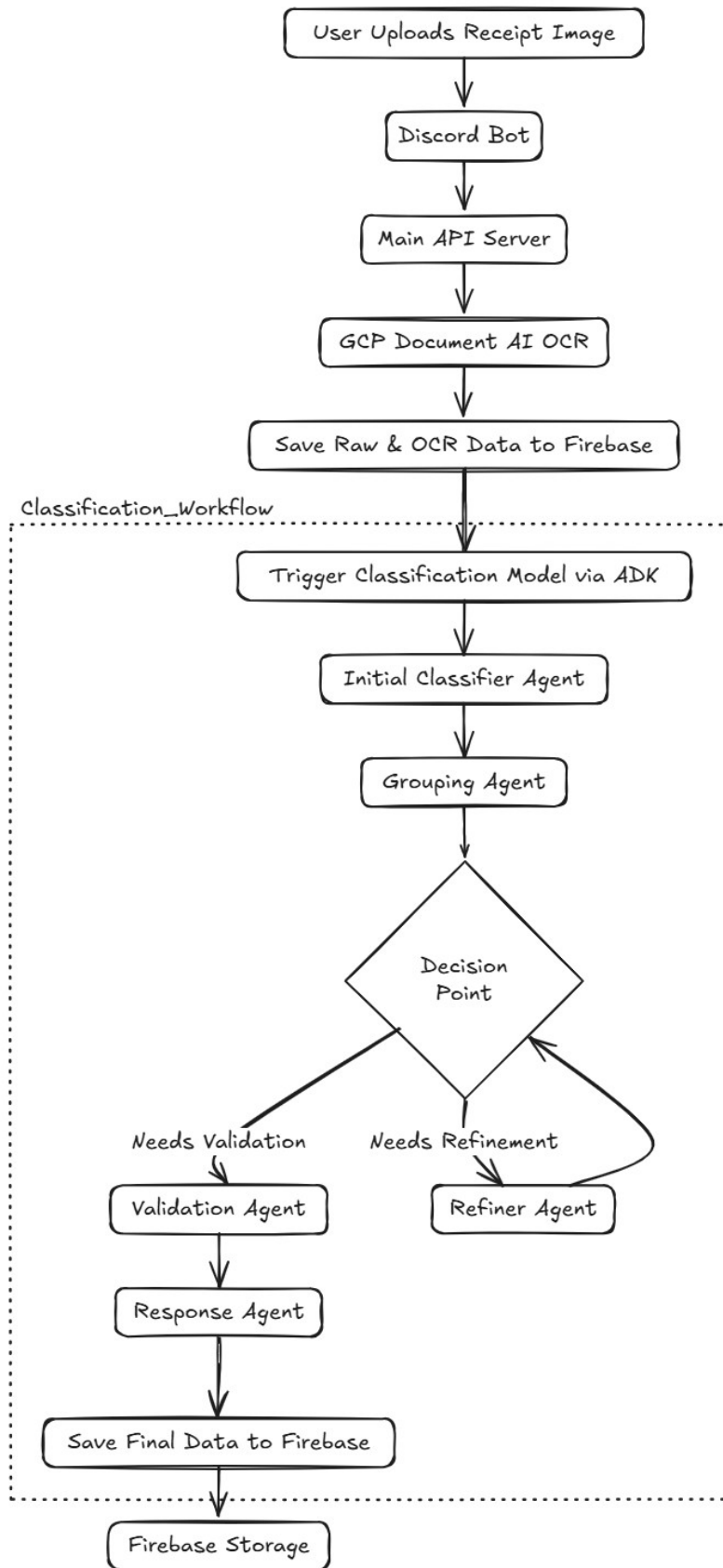
Flask, Python, Docker, Firebase, Google Cloud (Document AI, Cloud Run), ADK

Infrastructure

- Cloud Run containers for API and ADK
- VM hosting for Discord bot
- CI/CD automation using Dockerfiles and shell scripts

Data Flow & Pipeline Implementation

1. Image uploaded via Discord or web form.
2. Flask API routes the image to GCP Document AI.
3. OCR results sent to ADK classification pipeline.
4. ADK agents (Initial Classifier → Grouping Agent → Validation Agent → Finalizer) process and label data.
5. Results saved to Firebase under /RAW_DATA, /RECEIPTS, /SUMMARISED_DATA.
6. Dashboard fetches data from Firebase for visualization.



Testing & Evaluation

Testing Conducted

- Unit Tests: API routes, ADK module, classification functions.
- UAT: Discord bot tested by multiple users.
- Manual Tests: End-to-end pipeline validation from upload to classification.
- Model Validation: Holdout receipts tested for accuracy.

Results

- Classification accuracy robust for structured receipts.
- ADK pipeline more stable and explainable than previous LLM-based logic.
- OCR success rate: ~92% on clean receipt images.
- Bugs fixed: JSON formatting, Firebase writes, bot edge cases.

Challenges & Iterative Improvements

**Every time Jonathan broke the project*

- Challenge of Ollama Agent Inaccuracies
- Challenge faced because Lack of Examples and Documentations for ADK
- Development of the Flask API Logic so that it can be Integrated anywhere with a Secure Dashboard allowing Users from any Source to have a Single Login using Firebase OAuth2.0 Authentication.

Final Deployment & Demo

Deployment Status

- ADK Classification Pipeline: Cloud Run
- Flask API: Cloud Run
- Discord Bot: VM
- Firebase Storage
- Dashboard UI

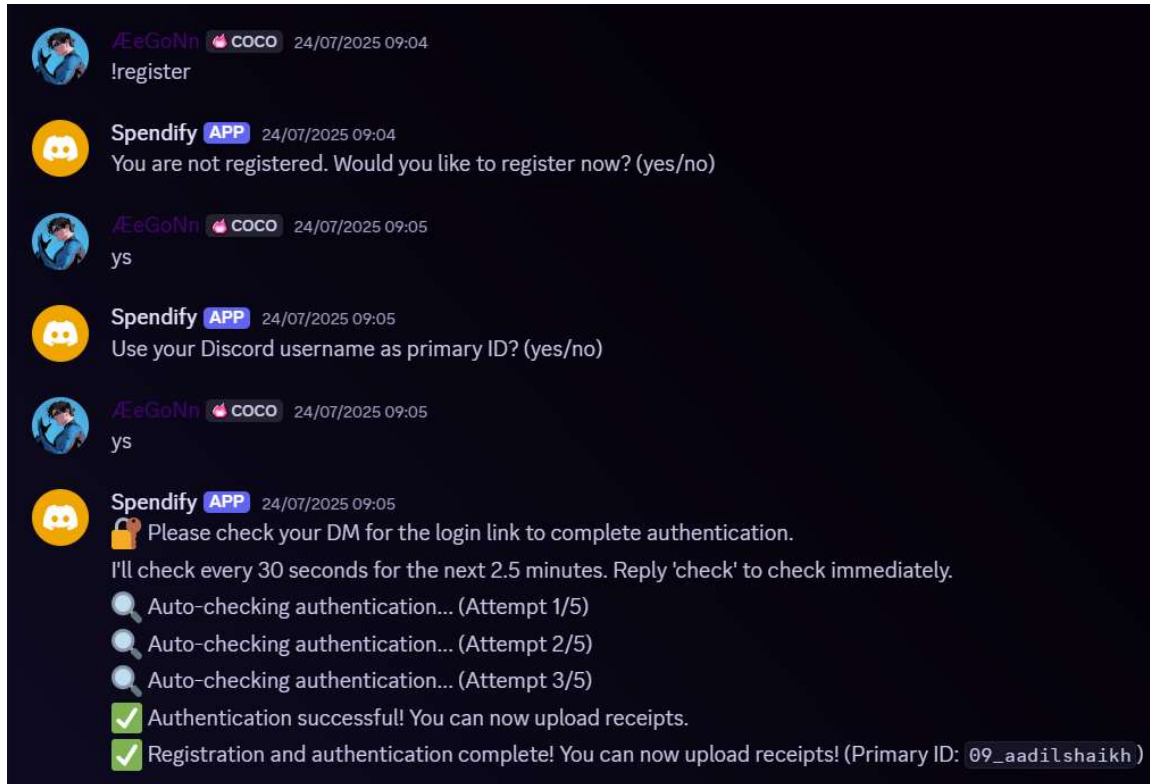
URLs

- ADK: <https://receipt-classifier-32377413295.us-central1.run.app/>
- API: <https://spendify-api-32377413295.us-central1.run.app/>
- Discord Bot:
https://discord.com/oauth2/authorize?client_id=1375145987106144297&permissions=8&scope=bot

Demo Summary

Users can upload a receipt via Discord and receive categorized, structured outputs within seconds. Future features will support full dashboard interactivity and spend prediction.

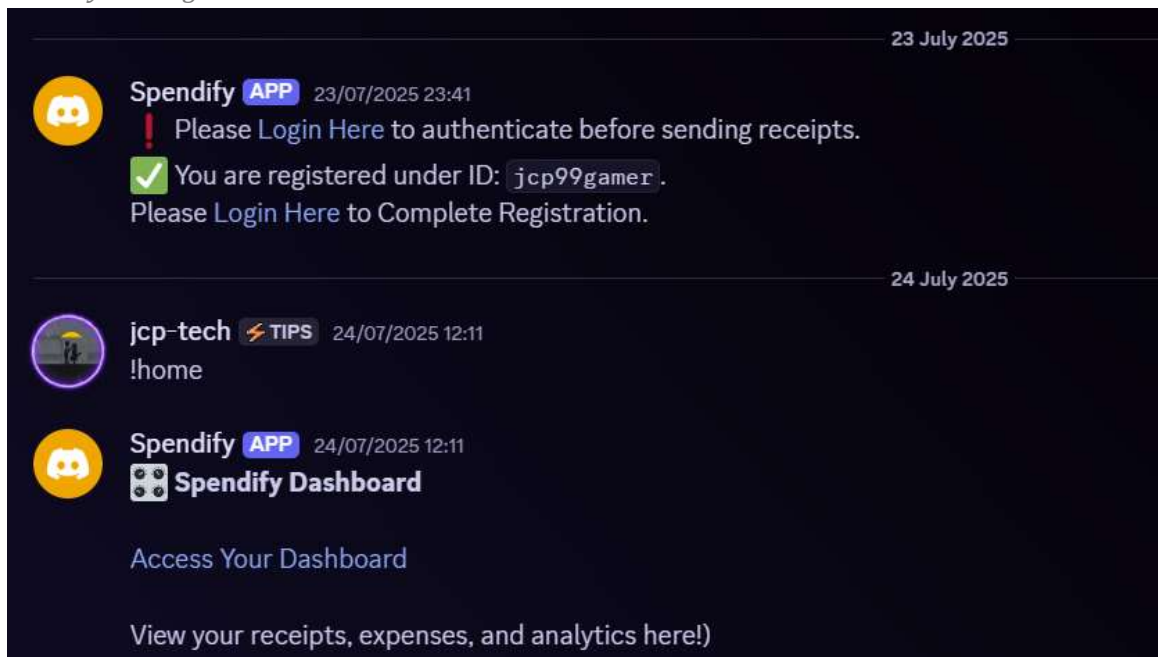
Registration Procedure (can be on a Server):



The screenshot shows a Discord chat interface with the following messages:

- AEeGoNn** (COCO) 24/07/2025 09:04: !register
- Spendify** (APP) 24/07/2025 09:04: You are not registered. Would you like to register now? (yes/no)
- AEeGoNn** (COCO) 24/07/2025 09:05: ys
- Spendify** (APP) 24/07/2025 09:05: Use your Discord username as primary ID? (yes/no)
- AEeGoNn** (COCO) 24/07/2025 09:05: ys
- Spendify** (APP) 24/07/2025 09:05:  Please check your DM for the login link to complete authentication. I'll check every 30 seconds for the next 2.5 minutes. Reply 'check' to check immediately.
- Spendify** (APP) 24/07/2025 09:05:  Auto-checking authentication... (Attempt 1/5)
- Spendify** (APP) 24/07/2025 09:05:  Auto-checking authentication... (Attempt 2/5)
- Spendify** (APP) 24/07/2025 09:05:  Auto-checking authentication... (Attempt 3/5)
- Spendify** (APP) 24/07/2025 09:05:  Authentication successful! You can now upload receipts.
- Spendify** (APP) 24/07/2025 09:05:  Registration and authentication complete! You can now upload receipts! (Primary ID: 09_aadilshaikh)

Directly Messaged User the Authentication Link:



The screenshot shows a Discord chat interface with the following messages:

- Spendify** (APP) 23/07/2025 23:41:  Please Login Here to authenticate before sending receipts.
- Spendify** (APP) 23/07/2025 23:41:  You are registered under ID: jcp99gamer.
- Spendify** (APP) 23/07/2025 23:41: Please Login Here to Complete Registration.
- jcp-tech** (TIPS) 24/07/2025 12:11: !home
- Spendify** (APP) 24/07/2025 12:11:  **Spendify Dashboard**
- Spendify** (APP) 24/07/2025 12:11: Access Your Dashboard
- Spendify** (APP) 24/07/2025 12:11: View your receipts, expenses, and analytics here!)

User Upload's an Image and the Spendify Bot Identifies and Processes it:



aliyyahj  02:09 PM





Spendify APP 02:09 PM
 Image saved, session `f d33b72d-5593-4e9b-8264-3d66ef82c136`. Uploading for processing...

ADK Classification showing the Event's Pipeline (Sample of what happens in Backend):

Agent Development Kit

receipt-classifier

Trace Events State Artifacts Sessions Eval

Conversations

0 text:{"classified": [{ "item": "g1 Imp White", "quan...

1 text:{"grouped": [{ "category": "Others", "items": [...

2 functionCall:calculate_final_total

3 functionResponse:calculate_final_total

4 functionCall:exit_function

5 functionResponse:exit_function

6 functionCall:save_to_firebase

7 functionResponse:save_to_firebase

8 text:''json { "result": "success", "message": "Data s...

SESSION ID 15598690-1636-4974-85d6-2210a250311e

Token Streaming + New Session

save_to_firestore

✓ save_to_firestore

```
{
  "result": "success",
  "message": "Data saved to Firebase.",
  "data": {
    "categories": [
      {
        "category": "Others",
        "items": [
          "g1 Imp White",
          "Blue Moon Tap"
        ],
        "total_price": "13.75"
      },
      {
        "category": "Fast Food",
        "items": [
          "Mozzarella&Tomato",
          "Pork Quesadilla",
          "Fren Onion Soup",
          "Pork Chop",
          "Hanger Sizzle"
        ],
        "total_price": "72.75"
      }
    ]
  }
}
```

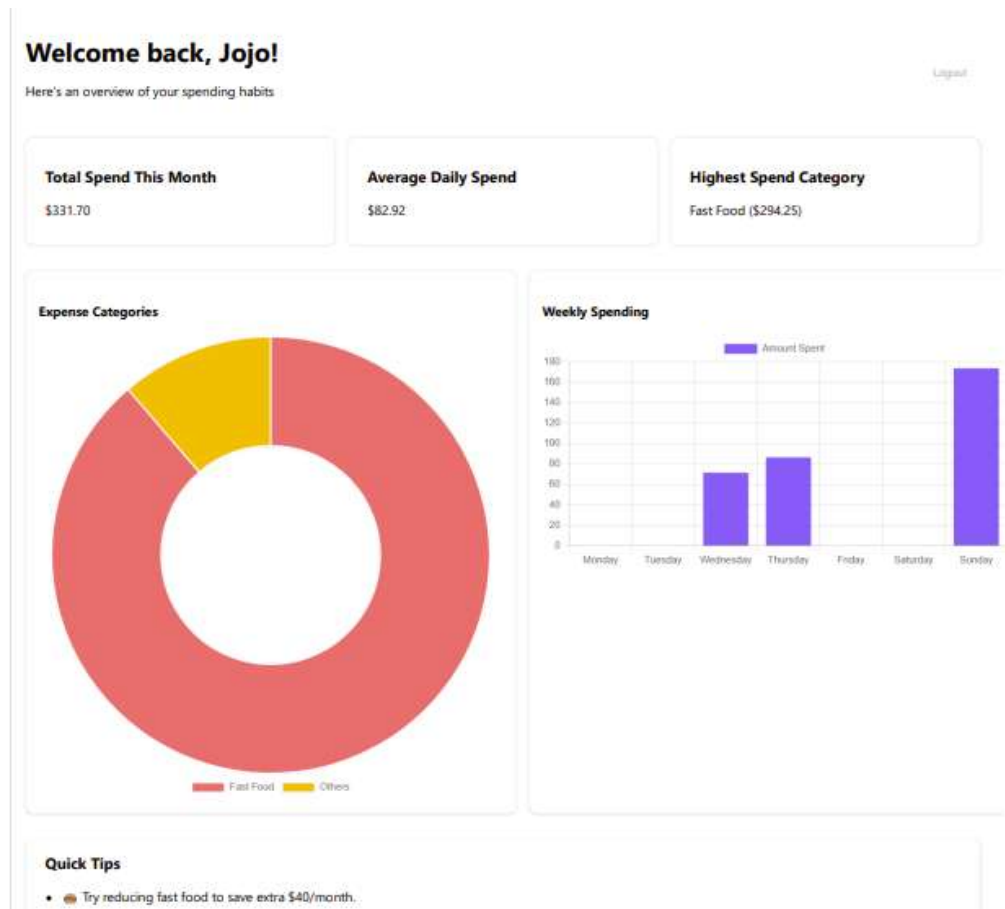
Type a Message...



[illegible]

The screenshot shows a web browser window with a dark theme. The address bar displays 'spendify-api-32377413295.us-central1.run.app'. The main content area is a light gray rectangle with the word 'Login' in bold black text and a blue 'Sign in with Google' button below it. A Google account selection overlay is visible on the right side of the screen. The overlay has a dark background and contains the text 'Choose an account to continue to spendify-mapple-masala.firebaseio.com'. It lists two accounts: 'Jonathan Chacko Pattasseril' with email 'jonathanchacko1805@gmail.com' and 'Jonathan Chacko' with email 'jonathan.chacko@maritimesupports.com'. At the bottom of the overlay, there is a link to 'Use another account'. The browser's status bar at the bottom shows 'English (United States)', 'Help', 'Privacy', and 'Terms' links.

User Dashboard:



User has Option to Upload Receipt from another Source like this Dashboard:

Spendify

- Dashboard
- Upload Receipt
- Chat with Bot

Upload Receipt

Select or drop an image file to process

[Logout](#)

Drag & drop your receipt image here, or click to select a file.

Ethical Considerations

Spendify processes financial documents that may include sensitive personal data. To uphold user privacy, all receipt images and processed data are securely stored in a private Firebase database, with access restricted to authenticated project components. No personal identifiers are logged or shared externally. The system is designed without user profiling or monetization at this stage to avoid misuse. Future expansions will include formal privacy policies and data consent mechanisms to align with ethical AI and data governance standards.

Conclusion & Future Work

Spendify successfully delivers an end-to-end receipt processing pipeline, replacing LLM fragility with robust agent workflows. With deployment in place and real-time Discord integration, the platform lays a strong foundation for production-ready tools.

Next Steps

- Finalize regression model integration
- Complete frontend dashboard features
- Implement real-time expense summaries
- Create a WhatsApp Bot for Spendify
- Demonstrate Implementation in Business System's like CRM's
- Add AI chat integration via it's Multiple Sources (eg: Discord and Dashboard)
- Explore monetization and scalability for productization

References

Discord. (n.d.). <https://discord.com/developers/docs/quick-start/getting-started#configuring-your-bot>

Flask. (n.d.). <https://flask.palletsprojects.com/en/stable/api/>

Firebase. (n.d.)

- Firestore= <https://firebase.google.com/docs/firestore/quickstart#python>
- Authentication= <https://firebase.google.com/docs/auth/web/google-signin>

Google Cloud. (n.d.). *Document AI overview*. <https://cloud.google.com/document-ai/docs/setup>

Google Cloud. (n.d.). *Agent Builder (ADK) documentation*. <https://google.github.io/adk-docs/>

AI Model Selection:

- Vertex AI Platform= <https://cloud.google.com/vertex-ai/generative-ai/docs/models>
- AI Studio= <https://ai.google.dev/gemini-api/docs/models>

GitHub. (2025). *Spendify project repository*. <https://github.com/jcp-tech/Spendify>